

RCA Clean - Standard set of wafer cleaning steps performed prior to high temp. processing steps (oxidation, diffusion, CVD) of Si wafers.

→ Werner Kern - 1965 - worked for RCA (Radio Corporation of America).

Chemical processes performed in sequence

- (i) Removal of organic contaminants (organic clean + particle clean)
- (ii) removal of thin oxide layer (oxide strip, optional)
- (iii) removal of ionic contamination (ionic clean)

Step 1 : SC-1 (Standard Clean)

• The first solution is a high pH soln consisting of

(i) 5 parts of deionized ~~water~~ water

(ii) 1 part of aqueous NH_4OH (29% by weight of NH_3)

(iii) 1 part of aqueous H_2O_2 (30%)

} $5\text{H}_2\text{O} : 1\text{NH}_4\text{OH} : 1\text{H}_2\text{O}_2$

• Wafers are placed in this solution at $70-80^\circ\text{C}$ for 10 minutes

- ✓ • The solution oxidizes organic films and complexes of Group IB & IIB metals and other metals like Au, Ag, Cu, Ni, Zn, Cd, Co and Cr.
- ✓ • Dissolves the thin native oxide layer on Si & grows a next oxide layer by oxidation.
- Combination of etching & reoxidation dislodges particles from the wafer surface.

Step 2: Oxide Strip (Optional)

- ↳ Bare Si wafers are dipped in 1:100 or 1:50 solⁿ of aqueous HF at 25°C for about 15s to remove the thin oxide layer & some fraction of ionic components.
- ↳ This step needs to be performed with ultra high purity materials & ultra clean containers, otherwise it can lead to recontamination (Si surface is very reactive)

Step 3: SC-2 or ionic clean

- ↳ SC-2 is a low pH solution consisting of $6H_2O : 1H_2O_2 (30\%) : 1HCl (37\%)$ in which the wafers are placed at $70-80^\circ C$ for 10 minutes.
- ↳ Solution removes alkalis and cations like Al^{3+} , Fe^{3+} and Mg^{2+} , that forms NH_4OH insoluble hydroxides in basic solutions like SC-1.
- ↳ Metals precipitating on the wafer surface in SC-1 solution form soluble complexes in SC-2 solution.
- ↳ Also removes metallic contaminants such as Au, which is not completely removed in SC-1 step.

Step 4: Rinsing & Drying

Performed with flowing water after RCA Clean -

Wafer surface submerged into water - easily recontaminated by organics or particulates floating on the surface of water



Gettering

- It is difficult to reduce the conc. of undesirable elements such as Fe, Cu, Au, etc and alkali ions Na^+ , K^+ , etc in Si materials to certain levels where they will not affect the device performance.

- Gettering is a third tier defense mechanism and is a process by which unwanted impurities are removed by providing an alternate location (getter) in regions of the chip where they ~~per~~ prefer to reside (lower energy regions) and do minimal harm.

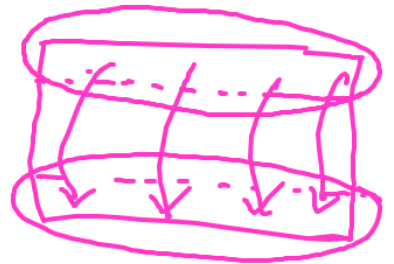
- Three step process

- (i) elements to be gettered are freed from the trapping sites they occupy & made mobile.

- (ii) they diffuse to the gettering site - undesirable elements

have high diffusivities & are easily captured in regions with mechanical defects or chemical traps.

- (iii) they must be trapped.



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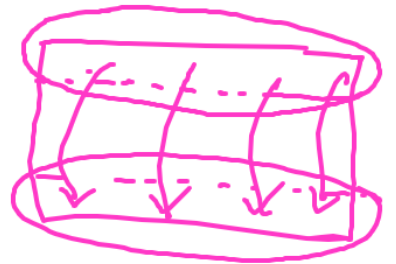
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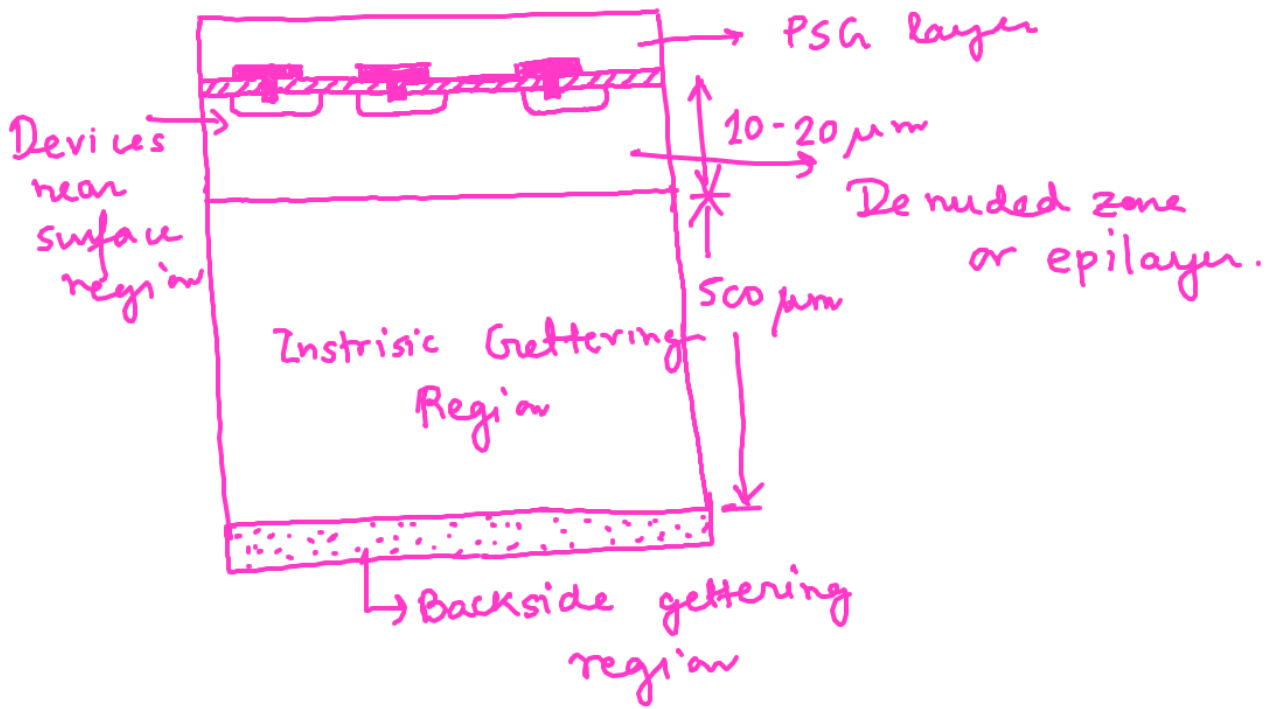
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PSG - Phosphosilicate glass
 $\text{P}_2\text{O}_5/\text{SiO}_2$ glass - deposited by CVD, LPCVD - an efficient trap for alkali ions - forms stable complex that binds Na^+ or K^+ ions.

Trace metal atoms in Si bulk are gettered by

(i) extrinsic gettering - use of a layer on wafer backside to trap trace atoms.

(ii) intrinsic gettering - CZ Si - as it contains oxygen. Oxygen is made to precipitate & create traps for metal atoms within the wafer bulk.